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The Late Miocene Carbon Isotope Shift driven by synergetic terrestrial processes: A box-model study

Jinlong Du [a,*], Jun Tian [a,*], Wentao Ma [b,c]

[a] State Key Laboratory of Marine Geology, Tongji University, Shanghai 200092, China
[b] State Key Laboratory of Satellite Ocean Environment Dynamics, Second Institute of Oceanography, Ministry of Natural Resources, Hangzhou, China
[c] Southern Marine Science and Engineering Guangdong Laboratory (Zhuhai), Zhuhai, China

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ABSTRACT

As the most significant carbon isotope excursion in the past ~10 Ma, the Late Miocene Carbon Isotope Shift (LMCIS, ~7.65 to 6.5 Ma) offers a great opportunity to investigate the carbon-climate dynamics in a warmer-than-today world. However, the driving mechanisms of the LMCIS remain controversial. In this study, we used a 7-box biogeochemical model to simulate the long-term sea water $\delta^{13}C$ and atmospheric CO_2 changes during the late Miocene. Based on quantitative parameterization of two terrestrial processes (C_4-grasses expansion and enhanced weathering input) during the late Miocene, our results show that the synergy between the two terrestrial processes may ultimately result in the LMCIS via the perturbation of the land-sea carbon fluxes. Moreover, our results reveal that the re-partitioning of alkalinity and nutrients between the land and the ocean may have influenced the long-term atmospheric CO_2 change during the late Miocene.

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1. Introduction

The late Miocene (~11.6 to 5.3 Ma) stands out as a period of considerable interest during the long-term Cenozoic cooling trend, as it is marked by profound climate changes in a much warmer world with modern-like atmospheric pCO_2 (pCO_2^{atm}) and land-sea configuration [see (Steinthorsdottir et al., 2021) for a thorough review]. In the late Miocene carbon cycle research, the mechanism driving the Late Miocene Carbon Isotope Shift (LMCIS, ~7.65 to 6.5 Ma) is one of the central questions, as the LMCIS could be the most notable $\delta^{13}C$ excursion in the past 10 Ma. First discussed in the early 1980s (Haq et al., 1980), the LMCIS was characterized by a profound negative shift of seawater stable carbon isotopic composition ($\delta^{13}C$) derived from deep-sea benthic foraminiferal archives (Westerhold et al., 2020) (Fig. 1A). During the LMCIS, $\delta^{13}C$ of bottom water ($\delta^{13}C_{bottom}$) is suggested to decrease by up to 0.9‰ (although the amplitude may vary in different basins). The deep-time carbon dynamics behind the LMCIS is intricately connected to tectonic/climatic events during the late Miocene. For instance, the late Miocene witnessed a profound decrease of sea surface temperatures (SST) at sites in almost all oceans (termed the Late Miocene Cooling, LMC, Herbert et al., 2016), with up to 8 °C cooling at high-latitude sites in both hemispheres. The pCO_2^{atm} reconstructions based on independent proxies (alkenone, diatom, foraminiferal boron isotope, and $\delta^{13}C$ of terrestrial C_3 plants) have shown a decline during the late Miocene, although absolute pCO_2^{atm} levels and amplitude of the pCO_2^{atm} decline are still controversial (Fig. 1C). The late Miocene saw also intensive uplift events in the northeast Tibetan Plateau (Fig. 1E), which has been reported to have driven the central-Asia aridification (Liu et al., 2016; Yang et al., 2019) and further linked to an evident expansion of C_4-grasslands at mid- to low-latitudes (e.g. Cerling et al., 1997, 1993; Feakins et al., 2020; Tauxe and Feakins, 2020). These tectonic/climatic changes are closely linked to each other and the late Miocene climate-carbon cycle dynamics surrounding the LMCIS, therefore, provides an opportunity to better understand the mechanisms functioning in a warmer-than-today world, which would help guide our predictions of future climate change.

However, the underlying mechanism of the LMCIS remains enigmatic, although a number of hypotheses have been proposed.

* Corresponding authors.
E-mail addresses: dujinlong@tongji.edu.cn (J. Du), tianjun@tongji.edu.cn (J. Tian).

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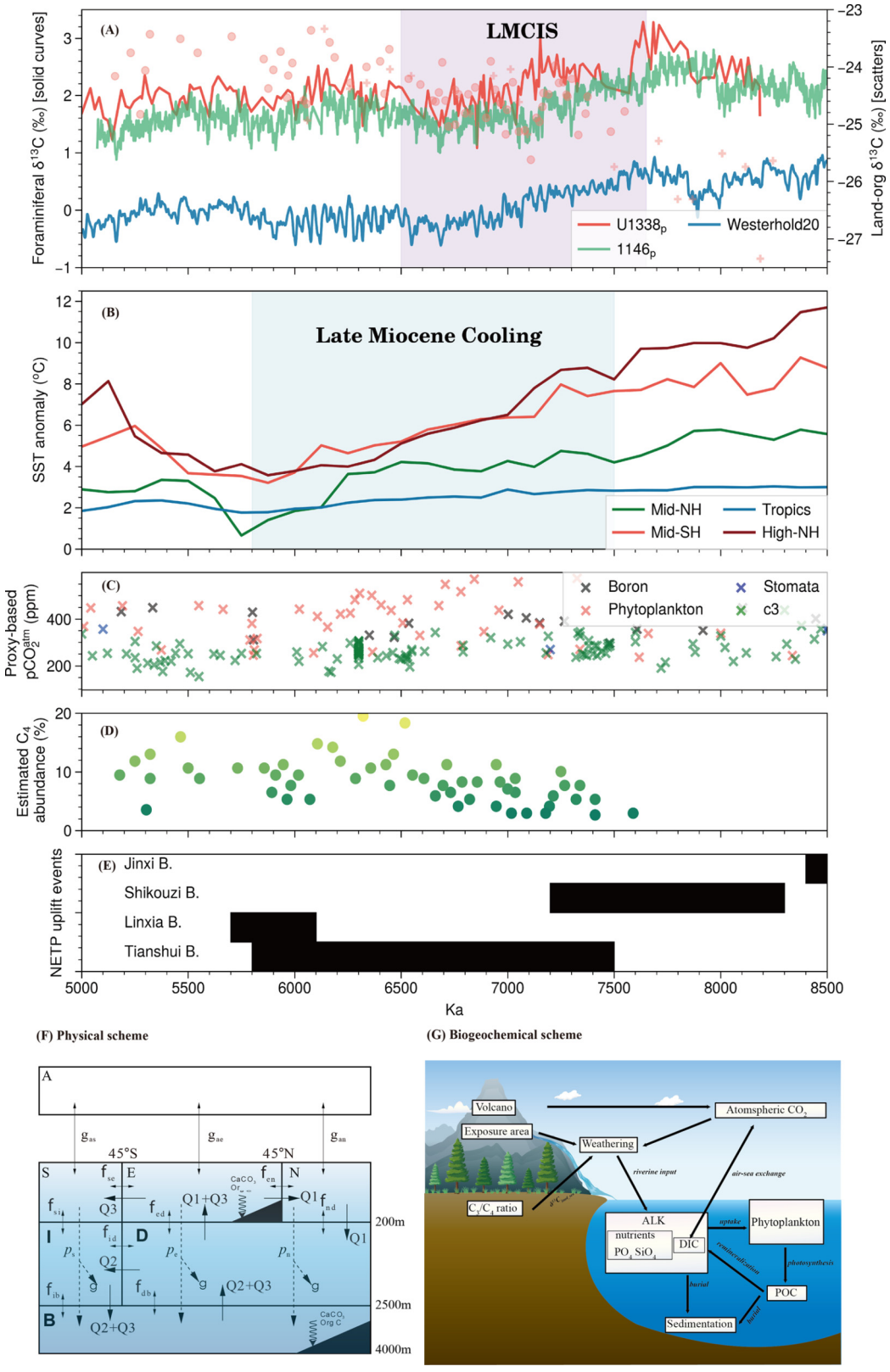


Fig. 1. Late-Miocene tectonic/climatic background and model architecture. (A) Marine foraminiferal $\delta^{13}C$ records (solid curves) and land-organic $\delta^{13}C$ records (gray scatters). In (A), planktonic $\delta^{13}C$ records at the Ocean Drilling Program (ODP) Site 1146 and Integrated Ocean Drilling Program (IODP) Site U1338 are from Holbourn et al. (2018) and Drury et al. (2018), respectively. "Westerhold20" is the composite benthic foraminiferal $\delta^{13}C$ record by Westerhold et al. (2020). Land-organic $\delta^{13}C$ records are from the Tianshui Basin ("+", Hou et al., 2015) and the Jiaxian section ("o", Gallagher et al., 2021) in China. (B) SST anomaly (compared to modern state) stacks from (Herbert et al., 2016). (C) Proxy-based late-Miocene pCO_2^{atm} estimation. Stomata-based data are from (Franks et al., 2014; Kürschner et al., 2008, 1996; Stults et al., 2011; Van Der Burgh et al., 1993); phytoplankton-based data are from (Bolton et al., 2016; Tanner et al., 2020); $\delta^{11}B$-based data are from (Guillermic et al., 2021; Sosdian et al., 2018); and C_3-$\delta^{13}C$-based data are from (Cui et al., 2020). (D) Estimated C_4 abundance from (Gallagher et al., 2021). (E) Northeast Tibetan Plateau (NETP) uplift events, adapted from (Miao et al., 2012). (F) Physical scheme used in this study. (G) Biogeochemical scheme used in this study. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

One of the prevailing hypotheses is that the enhanced marine primary productivity (PP) during the late Miocene has driven the LMCIS (Grant and Dickens, 2002; Haq et al., 1980). As photosynthesis tends to utilize ^{12}C first, greater primary production would lead to a negative $\delta^{13}C_{bottom}$ shift and an enlarged vertical $\delta^{13}C$ gradient. Nevertheless, the enhanced PP during the late Miocene tends to be regional, mainly reported at the Southern Ocean (Diester-Haass et al., 2005; Grant and Dickens, 2002) or the equatorial Pacific (Lyle et al., 2018). In addition, enhanced PP linked to the late Miocene-early Pliocene biogenic bloom event lasts much longer than the LMCIS (Diester-Haass et al., 2005; Lyle et al., 2018). The other hypothesis stresses the reorganization of ocean circulation (and/or a perturbation in the source-water $\delta^{13}C$), which changes deep water $\delta^{13}C$ by altering the fraction between northern- and southern-sourced deep waters in the deep ocean water mixing process (Butzin et al., 2011; Hodell and Kathryn, 2006). This hypothesis is (partly) supported by the inter-basin $\delta^{13}C$ gradient (Poore et al., 2006). However, a modeling study has tested the impact of possible circulation changes on $\delta^{13}C_{seawater}$ (Butzin et al., 2011), but did not capture the negative $\delta^{13}C$ shift in the deep Atlantic. Moreover, changes in the source-water $\delta^{13}C$ would be merely hypothetical, as the source-water $\delta^{13}C$ during the late Miocene is poorly constrained.

Without external carbon (isotopic) source/sink, the above hypotheses implicitly assumed an enhanced vertical/inter-basin $\delta^{13}C$ gradient. These mechanisms, however, are challenged by planktonic foraminiferal $\delta^{13}C$ ($\delta^{13}C_p$) records exhibiting similar decreasing trend as the benthic foraminiferal $\delta^{13}C$ (Drury et al., 2018; Holbourn et al., 2018) (Fig. 1A). The remarkable similarity of the deep and surface oceanic $\delta^{13}C$ during the LMCIS implicates that a negative $\delta^{13}C$ source may have played a key role in driving the LMCIS of global oceans.

Holding a large amount of carbon reacting on long timescales (over ~100 ka, Hülse et al., 2017), the terrigenous carbon is the most likely source. Two documented terrestrial processes may contribute to the LMCIS. The first is the transportation of terrigenous carbon via weathering, which could produce a negative oceanic $\delta^{13}C$ shift as proposed by Bickert et al. (2004). Intense weathering can transfer ^{13}C-depleted carbon (Sanyal et al., 2005) into the ocean. This hypothesis could be indirectly supported by a reverse model study, showing that the continental weathering (both carbonate and silicate weathering) has been enhanced in the past 10 Ma (Li and Elderfield, 2013). The other terrestrial process is a major transition of the terrestrial ecosystems during the late Miocene, featuring an evident expansion of C_4 grasslands at the expense of C_3-dominated forest at mid- to low-latitudes (e.g. Cerling et al., 1997, 1993; Feakins et al., 2020; Fox et al., 2012; Hoetzel et al., 2013; Quade and Cerling, 1995; Tauxe and Feakins, 2020). The C_4 photosynthetic pathway is associated with a much reduced isotopic preference for ^{12}C (Farquhar et al., 1989), resulting in the different $\delta^{13}C$ between C_3-plants (~-27‰) and C_4-grasses (~-13‰). One of the direct effects of C_4-expansion, therefore, would be a clear increase in $\delta^{13}C$ of the terrigenous organic matters ($\delta^{13}C_{land,org}$). For instance, as shown by Hou et al. (2015), the late Miocene witnessed a ~2.4‰ increase of $\delta^{13}C_{land,org}$ from 8 to 6 Ma in the Tianshui Basin. As a larger fraction of ^{13}C was sequestered in the land biomass during the C_4-expansion, more ^{12}C would be redistributed to the atmosphere and the ocean, if the total amount of carbon in the land-sea-air system remained unchanged (which, however, needs further validation). According to Tipple et al. (2010), $\delta^{13}C_{CO2atm}$ fluctuated around -6.5‰ during the late Miocene and did not show a drastic decrease, leaving the ocean as the most plausible ultimate recipient of the ^{12}C redistribution. The land-sea ^{12}C transportation was likely to be done via surface run-off; and we based some of our experimental design (described later) on this assumption.

Modeling has been used to test the impact of terrestrial processes during the LMCIS. For instance, based on a simplified mass-balance model study, Kump and Arthur (1999) suggested a ~1‰ decrease of $\delta^{13}C_{seawater}$ with 50% C_4-grasses in the land biome. In contrast, Diester-Haass et al. (2006) showed that the influence of C_4-expansion might be trivial (0.1‰), and the carbon transferred into the deep ocean played the key role (over 0.6‰). The inconsistency between these modeling studies shows that a comprehensive study is still needed to coordinate the individual and the synergistic role of terrestrial processes quantitatively. In this study, we provide our model results regarding the terrestrial forcing on the LMCIS using a 7-box model. Both seawater $\delta^{13}C$ and pCO_2^{atm} are simulated in our study. Based on a control run and three sensitivity experiments, we suggest a controlling role of terrestrial forcing in the full course of the LMCIS.

2. Model description and experimental design

2.1. Description of the 7-box model

While detailed description and parameterization of the model can be found in the Supplementary Material, we here describe the essential features of our model for better readability. The 7-box biogeochemical model used in this study is adapted from a previous research which focused on the mid-Miocene carbon cycle (Ma et al., 2011). The model contains one atmospheric box and six oceanic boxes (Fig. 1F). The prescribed physical framework was calibrated for better representing the late Miocene condition following the results from previous fully-coupled climate model studies (Knorr et al., 2011; Stein et al., 2016). In our model, the atmospheric box receives CO_2 from volcanic and kerogen degassing and exchanges CO_2 with the three surface boxes based on the air-sea pCO_2 difference. Divided by 45° N/S, the surface ocean in our model contains a northern box "N", an equatorial box "E", and a southern box "S". The deep ocean contains a deep box "D", a southern intermediate box "I", and a bottom box "B". Essential tracers including phosphate, dissolved inorganic carbon (DIC), alkalinity and $\delta^{13}C$ are transported and mixed in all six ocean boxes. The equatorial box "E" receives terrigenous carbonate, silicate, phosphate, and hence alkalinity via runoff. Primary production takes place in the three surface boxes, while phosphate is taken as the limit nutrient. Carbon is removed by the burial of $CaCO_3$ and organic matter, but carbon can also re-enter the ocean via $CaCO_3$ dissolution. In addition to the biogeochemical scheme described by Ma et al. (2011), we used a C_3-C_4 module in which $\delta^{13}C$ of riverine carbon is influenced by the fraction of C_4-plants within land ecosystems to depict land-sea redistribution of carbon isotopes due to C_4-expansion (Fig. 1G).

The air-sea carbon system in the model is driven by the weathering fluxes, which are further linked to the ETP parameter (i.e., the sum of normalized eccentricity, obliquity, and precession in a ratio of 1: 1: -1) (Laskar et al., 2004) and a tectonic-controlled perturbation term (Eq. S1); the $\delta^{13}C$ of the land-sea riverine flux is determined by the fraction of C_4-grasses contribution to the terrestrial primary production. A detailed list of parameters used in this study and the steady state of the model can be found in Table S1, S2 and S3.

2.2. Experimental design

In each experiment, the model was spun up for 4 Ma under the late Miocene boundary condition derived from a fully coupled model study (Knorr et al., 2011). The results in the spin-up stage generally agree well with the pre-LMCIS records (Table S3). Then, we designed and conducted four runs to investigate the role of the terrestrial forcing on the LMCIS as follows.

CNTR refers to the control run, which represents the land-sea-air carbon cycle under the Late Miocene orbital forcing. In the CNTR experiment, the system was directly driven by Earth's orbital parameters (ETP, which is the sum of normalized eccentricity, obliquity, and precession in a ratio of 1: 1: -1) (Laskar et al., 2004). The CNTR experiment is used as a reference for further comparison. Note in the control and following experiments, the initial and the weathering-referenced $pCO_2{}^{atm}$ was set to the Pre-Industrial (PI) level (280 ppm). This may be a modest estimation of the late Miocene $pCO_2{}^{atm}$ level as most proxies indicate roughly 300∼350 ppm at the late Miocene (Steinthorsdottir et al., 2021); however we chose to use this value for several reasons. First, while the major focus of this study is the $\delta^{13}C$ shift during the LMCIS, the terrestrial C_3-$\delta^{13}C$-based $pCO_2{}^{atm}$ reconstruction (Cui et al., 2020) shows a value of ∼280 ppm at ∼8 Ma (although the proxy itself is now being questioned because it does not counter for water stress). The ∼280 ppm would be a reasonable choice, taking the implicit connection of the land-sea ^{13}C redistribution in the context (this will be discussed later). Second, the 280 ppm was chosen by several previous studies from which we derived the late Miocene physical framework (e.g. Knorr et al., 2011), so we believe it would be better to align with the original setups.

VEG refers to the C_4-expansion experiment. As the fraction of C_4-grasses in the land biome is poorly constrained, we tuned a time-dependent parameter (see the Supplementary Material) to assume that the fraction stabilized at 2% before 8 Ma, increased from 2% to 20% logarithmically during 8 to 6 Ma. This assumption was based on studies showing that C_4-grasses contribute roughly ∼25% of the terrigenous primary production (Still et al., 2003) but the number should be rather low (or close to zero) before the late Miocene (Gallagher et al., 2021) (Fig. 1D).

LAND refers to the enhanced land input experiment. As there is no high-resolution reconstruction of the excess carbon transferred from land to ocean, the best available data are from the paleoceanographic reconstructions showing an exponentially growing number of terrigenous sediments deposited on the seafloor since Miocene (Hay et al., 1988; Zhang et al., 2001). Therefore, we applied a best-fit approach in this case. We first took the sediment amount at 8 Ma in the paleoceanographic reconstruction (Zhang et al., 2001) as a reference number ($Burial_{deep,ref}$) and then calculated the ratio between deep burial given in the same study and $Burial_{deep,ref}$. The ratio grows exponentially to 1.06 at ∼6 Ma, as shown in Fig. 2(A). Then we tuned the perturbation term ψ_i (Eq. S1) in our model until the deep carbonate burial in our model shows the same trend (with 1 at 8 Ma and 1.06 at 6 Ma). We chose to use the growth rate instead of the absolute value because of the scarcity and uncertainties of accessible archives.

INT refers to an integrated test, in which both the terrestrial forcings in the VEG and LAND experiments were applied. The INT experiment is used to examine the possible synergistic interplay of VEG and LAND, and we suggested a new hypothesis from the results in the INT experiment.

3. Results

3.1. Control and sensitivity experiments

We first conducted a control run (CNTR) as a reference experiment, in which the system was driven solely by the ETP parameter without any other perturbation. In this case, both $\delta^{13}C_{surface}$ (Fig. 4B, dashed yellow line) and $\delta^{13}C_{bottom}$ (Fig. 4B, solid yellow line) in the CNTR experiment show strong 405-kyr cycles, while no significant long-term shift of $\delta^{13}C_{surface}$ or $\delta^{13}C_{bottom}$ is observed, as can be seen in the 1-Myr LOESS mean $\delta^{13}C$ (Fig. 4C, yellow lines). The prominent ∼405-kyr cycles in $\delta^{13}C_{bottom}$ have been proposed as the "heartbeat" of the earth system (Pälike et al., 2006) and are widely found in deep-sea archives over the whole Cenozoic history (Westerhold et al., 2020). Though the significant 405-kyr cycles in the oceanic $\delta^{13}C$ simulated in the CNTR experiment highlight the importance of orbital forcing on the global carbon budget, the absence of long-term $\delta^{13}C$ shift in the CNTR experiment has excluded orbital forcing from being the primary driver of the LMCIS.

In the first sensitivity run VEG, we assumed that the fraction of C_4-grass in the land plants biome increased from 2% to 20% during 8 to 6 Ma and then stabilized at 20%. In this case, more ^{13}C would be sequestrated within the land biome, as C_4-grasses are significantly more ^{13}C-enriched than C_3-plants. The terrestrial ^{13}C sequestration would result in more ^{12}C transferring into the ocean via ^{13}C-depletion of the land-sea carbon flux, indicating that $\delta^{13}C$ of the land-sea carbon flux ($\delta^{13}C_{land_sea}$) should be anti-correlated with $\delta^{13}C_{land,org}$ during the C_4-expansion (as shown in Fig. 3). According to our assumption (fraction of C_4-grasses from 2% to 20%), $\delta^{13}C_{land_sea}$ in the VEG experiment decreases by ∼1.2‰ from 8 to 6 Ma (Eq. S10; Fig. 3). The decrease of $\delta^{13}C$ of land-sea carbon flux served then as the forcing in the VEG experiment. In the VEG experiment, the modeled $\delta^{13}C_{surface}$ and $\delta^{13}C_{bottom}$ show a paralleled long-term decrease of ∼0.26‰ from ∼7.8 Ma to 6.1 Ma (Fig. 4C, blue lines). The modeled decrease of sea water $\delta^{13}C$ is much lower than that in a well-known study (Kump and Arthur, 1999), in which the authors argued that if the fraction of C_4-plants reached 50%, $\delta^{13}C$ of sea water might decrease by ∼1‰. We ascribe the inconsistency to two reasons. First, in the Kump and Arthur's model, the authors assumed C_4-grasses contributed 50% of the terrestrial primary production, which may be an overestimation as today C_4-grasses contribute only about 25% (Gallagher et al., 2021; Still et al., 2003). Second, the baseline weathering flux in their model (34×10^{12} mol C/yr) is much larger than the baseline carbonate weathering flux in our model (12×10^{12} mol C/yr, Table S2). Beyond the $\delta^{13}C$ value of land-sea carbon transfer, the larger flux in their model may more significantly influence the $\delta^{13}C_{seawater}$ by more ^{12}C transportation into the ocean.

The second sensitivity run, LAND, is used to assess the effects of enhanced land-to-ocean input during the late Miocene. Previous studies have shown that the terrigenous carbonate deposited on the sea floor may increase exponentially by 6% from 8 to 6 Ma (Hay et al., 1988; Zhang et al., 2001), indicating more riverine weathering carbon input during the late Miocene. Additionally, in a more recent study, Caves et al. (2016) calculated the long-term carbonate weathering flux to increase ∼5% from 8 to 6 Ma. As the weathering flux is more ^{13}C-depleted than sea water (Sanyal et al., 2005), the enhanced weathering input would lead to negative $\delta^{13}C$ shift of sea water via mixing and diffusion. However, so far there seems to be no high-resolution reconstruction of the weathering flux during the LMCIS; hence it is difficult to determine accurate amount of the enhanced weathering input in the LAND experiment. We note that it may not be the best solution to simply assume that weathering input increases by 6% from 8 to 6 Ma, because the deep-sea carbonate burial is not linearly correlated with the total weathering input due to marine carbonate chemistry. Therefore, we chose to use the best-fit approach to constrain the total excess terrestrial input of carbon in the LAND experiment. This is done by tuning the parameter ψ_i in Eq. S1 (see the Supplementary Material), which determines the excess weathering input in our model. Different weathering input results in variable deep carbonate burial, and the target is a 6% exponential increase of deep carbonate burial to match the estimated growth rate of terrigenous sediment in previous studies (Hay et al., 1988; Zhang et al., 2001) (Fig. 2, AB). Compared with the CNTR experiment, we constrain the excess carbon transferred from land to sea during 8 to 6 Ma to be ∼2100 Gt, which further served as the forcing in the LAND experiment. Note in the LAND experiment, $\delta^{13}C_{land_sea}$

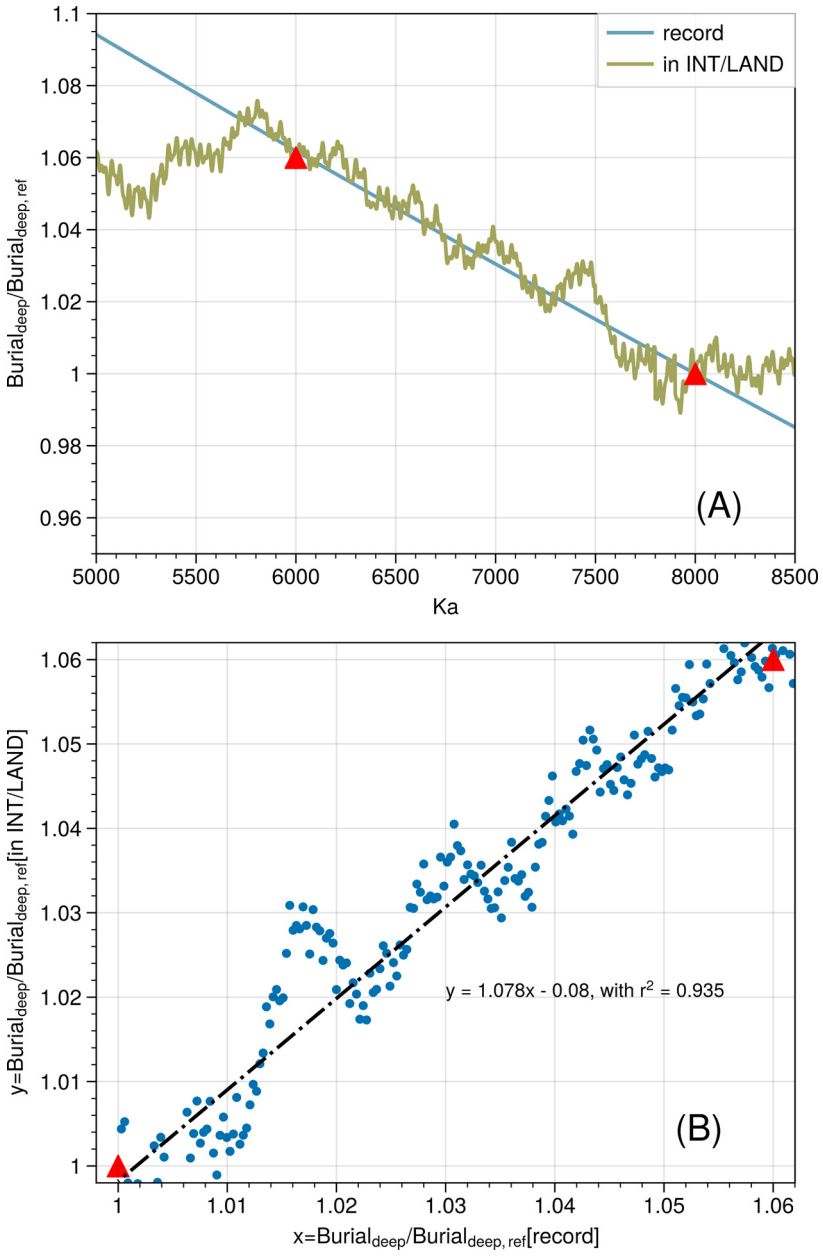


Fig. 2. Forcing used in the LAND experiment. (A) Comparison of the deep carbonate burial with respect to reference deep carbonate burial at 8 Ma in LAND/INT and Zhang et al., 2001. (B) Scatter plot of the two curves in (A). Note the deep carbonate burial in INT/LAND is considerably in line with that in the previous estimation. In (A)(B), the two red triangles reflect the state at 8 Ma and 6 Ma, respectively.

was the same as that in the CNTR experiment (-3.2‰) and did not change. As a result, both the modeled long-term $\delta^{13}C_{surface}$ (Fig. 4C, dashed cyan line) and $\delta^{13}C_{bottom}$ (Fig. 4C, solid cyan line) in the LAND experiment exhibit a decrease of ~0.27‰ during 7.6 to 6.8 Ma and then a gradual increase from 6.8 to 5.5 Ma. Because of the gradual increase, both the modeled $\delta^{13}C_{surface}$ and $\delta^{13}C_{bottom}$ in the LAND experiment are higher than those in the VEG experiment (Fig. 4C, cyan lines vs. blue lines) at the end of the simulations. We also note that the long-term trends of the modeled $\delta^{13}C_{surface}$ and $\delta^{13}C_{bottom}$ in the LAND experiment are similar but not paralleled, as the modeled $\delta^{13}C_{surface}$ increases slightly faster than $\delta^{13}C_{bottom}$ from 6.8 Ma on.

Overall, results from the sensitivity tests show that terrestrial forcing, either derived from C_4-expansion or enhanced weathering input, can introduce clear negative $\delta^{13}C$ shift of sea water. However, the long-term decreases in the oceanic $\delta^{13}C$ during the LMCIS in the VEG and LAND experiments (no more than 0.3‰) are too small to be consistent with that of the benthic and planktonic foraminiferal $\delta^{13}C$ records, indicating that a single terrestrial process could not trigger the LMCIS. Based on this, we further conducted an integrated run (INT) driven by the combined forcings in the VEG and LAND experiments.

3.2. The INT experiment and model-data comparison

The integrated run, INT, is used to assess the effect of combining terrestrial forcing in the VEG and LAND experiments (i.e., the C_4-expansion and the enhanced weathering input). The modeling results from the INT experiment clearly show that both the modeled $\delta^{13}C_{bottom}$ (Fig. 4B, solid green line) and its long-term trend (Fig. 4C, solid green line) agree reasonably well with the benthic foraminiferal $\delta^{13}C$ record (Fig. 4B, purple asterisks; Fig. 4C, solid

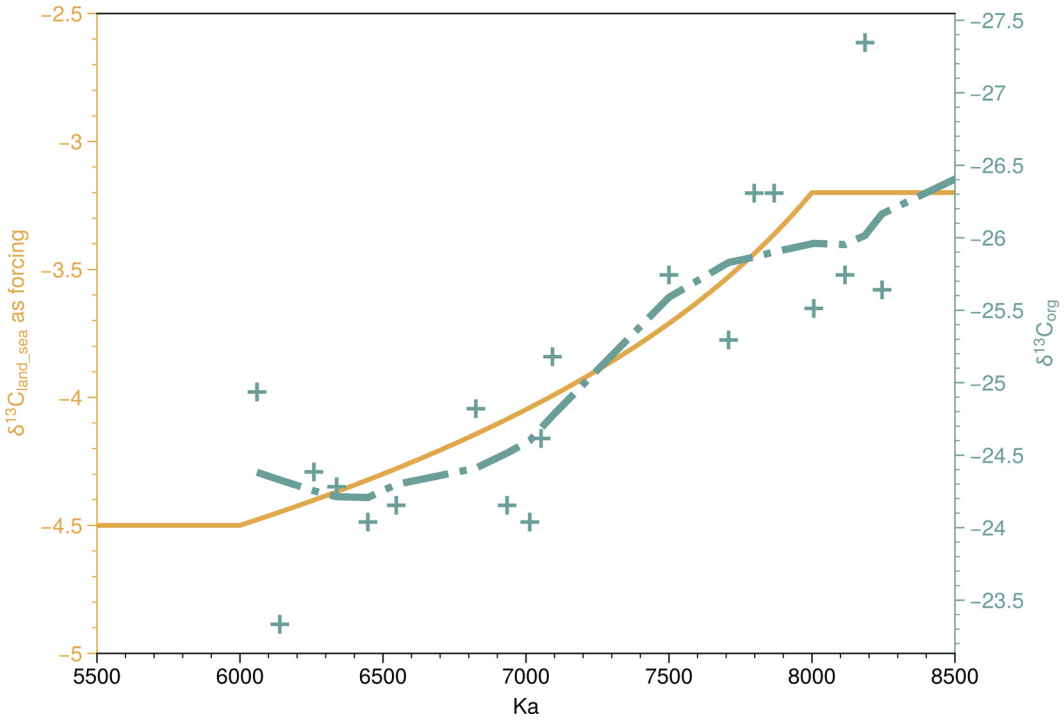


Fig. 3. Forcing used in the VEG experiment. The yellow solid line is $\delta^{13}C_{land_sea}$ used in VEG/INT. The green dashed line and "+" are archived $\delta^{13}C_{land,org}$ from (Hou et al., 2015).

purple line). Starting from ~7.6 Ma, long-term $\delta^{13}C_{bottom}$ in the INT experiment decreases by 0.72‰ in the following ~1.1 million years. During the same time period (7.6 to 6.5 Ma), the long-term $\delta^{13}C_{surface}$ in the INT experiment (Fig. 4C, dashed green line) also decreases by 0.63‰. Notably, the long-term decrease of the modeled $\delta^{13}C$ of either bottom or surface water in the INT experiment is not simply the sum of the modeled long-term $\delta^{13}C$ decreases in the VEG and LAND experiments, but significantly larger, indicating a synergistic interplay of the two forcing mechanisms (C_4-expansion and enhanced weathering input).

We further compare the modeled $\delta^{13}C_{surface}$, $\delta^{13}C_{bottom}$, and $pCO_2{}^{atm}$ in the INT experiment with recent-published paleoceanographic reconstructions (Cui et al., 2020; Drury et al., 2018; Holbourn et al., 2018). Generally, the modeled $\delta^{13}C_{surface}$ in the INT experiment (Fig. 5A, dark green) agrees well with the planktonic foraminiferal $\delta^{13}C$ at both the ODP Site 1146 (Fig. 5A, light green) and the IODP Site U1338 (Fig. 5A, olive), although long-term decrease of the modeled $\delta^{13}C_{surface}$ seems to be ~0.1 Ma later than that of the planktonic foraminiferal $\delta^{13}C$. Clearly, the modeled $\delta^{13}C_{bottom}$ is well in line with the composite benthic foraminiferal $\delta^{13}C$ record (Fig. 5A, dark blue vs. light blue), in terms of both the long-term trend and the short-term variation. In the INT experiment, the vertical $\delta^{13}C$ gradient (i.e., the $\delta^{13}C$ difference between the $\delta^{13}C_{surface}$ and $\delta^{13}C_{bottom}$; Fig. 5A, black dashed line) is slightly higher by 0.15‰ before 7.2 Ma than after 7.0 Ma. In general, the modeled $\delta^{13}C$ gradient of the deep and upper ocean shows very little variability from 8.5 Ma to 5 Ma, being consistent with the paleoceanographic records, such as the gradient between the planktonic and benthic foraminiferal $\delta^{13}C$ at ODP Site 1146 (Holbourn et al., 2018).

In the INT experiment, our results show that modeled late Miocene $pCO_2{}^{atm}$ fluctuates around an average of ~290 ppm before the LMCIS (Fig. 5B, brown). Under terrestrial forcing, modeled $pCO_2{}^{atm}$ in the INT experiment shows a long-term decrease of ~50 ppm from 8 to 5.8 Ma (Fig. 5C, brown). The drastic decrease starting from ~7.4 Ma (Fig. 5C, brown) is slightly later than the onset of the LMCIS (~7.6 Ma in our case). The modeled late Miocene $pCO_2{}^{atm}$ (brown curves in Fig. 5B and 5C) are in line with a recent paleo-$pCO_2{}^{atm}$ reconstruction (the thick black dashed line in Fig. 5B and the black curve in Fig. 5C) using $\delta^{13}C$ of terrestrial C_3 plants (Cui et al., 2020). We chose to compare our results with this C_3-$\delta^{13}C$-based reconstruction because the long-term change of the modeled $pCO_2{}^{atm}$ and $\delta^{13}C$ are linked to each other in our model by the weathering flux which redistributes carbon (and carbon isotopes as well) between the land and the ocean, while $\delta^{13}C$ signal of land C_3-plants gives rise to the $pCO_2{}^{atm}$ estimation by Cui et al., which, in turn, makes our model results comparable to their estimation. We note that it remains highly challenging to investigate the late Miocene $pCO_2{}^{atm}$ due to the scarcity of deep-time sedimentary records and the uncertainties in the proxies themselves. The C_3-$\delta^{13}C$-based proxy itself, for example, has been questioned because the proxy does not take into account water stress and may lead to unbound results (Kohn et al., 2015; Konrad et al., 2021; Lomax et al., 2019; Porter et al., 2019). However, based on multi-proxy reconstructions, the late-Miocene $pCO_2{}^{atm}$ was likely to be around ~250-350 ppm, which is close to (or slightly lower than) the modern level. At the aspect of terrestrial paleo-$pCO_2{}^{atm}$ proxies, for instance, terrestrial stomatal proxy reconstructions record $pCO_2{}^{atm}$ of ~350 ppm in the late Miocene (Franks et al., 2014; Kürschner et al., 2008, 1996; Stults et al., 2011; Van Der Burgh et al., 1993), while paleosol-derived $pCO_2{}^{atm}$ reconstructions show a lower value of ~260-280 ppm (Breecker and Retallack, 2014; Ji et al., 2018). Due to the low temporal resolution of terrestrial proxies, it is rather difficult to infer the long-term trend of $pCO_2{}^{atm}$ changes during the LMCIS. On the other hand, high-resolution marine paleo-$pCO_2{}^{atm}$ reconstructions, including alkenone (Bolton et al., 2016; Tanner et al., 2020), diatom-derived (Mejía et al., 2017) and foraminiferal $\delta^{11}B$ proxies (Guillermic et al., 2021; Sosdian et al., 2018), have revealed a $pCO_2{}^{atm}$ decline during the late Miocene. Although the absolute amplitude of the decline may vary from ~300 ppm to ~100 ppm depending on the proxy, a 30% draw-down of $pCO_2{}^{atm}$ is expected

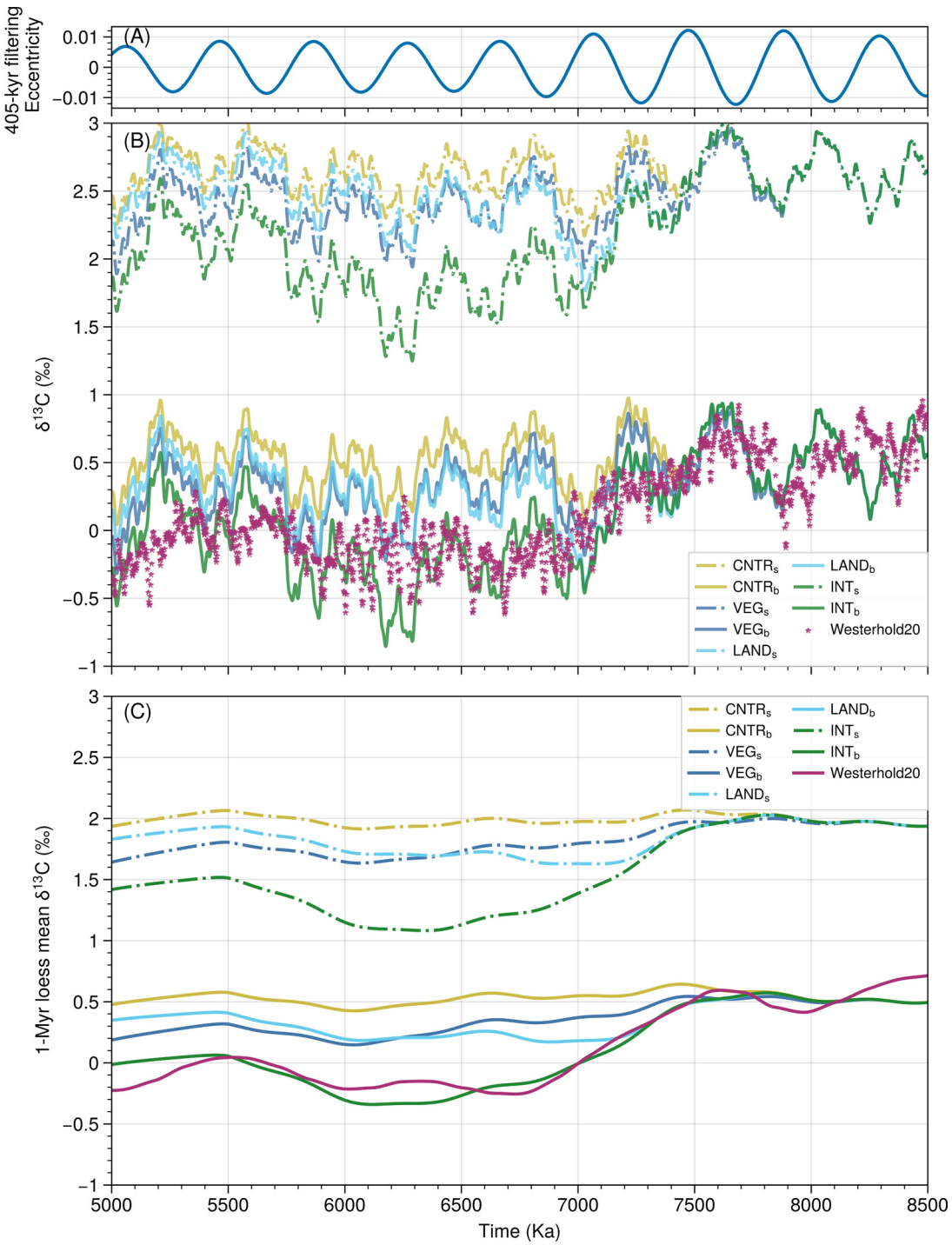


Fig. 4. (A) 405-kyr bandwidth filtering of standard Eccentricity; (B) Benthic foraminiferal $\delta^{13}C$ in a composite record (purple asterisks, from Westerhold et al., 2020) and bottom $\delta^{13}C$ our simulations (solid lines); (C) Comparison of the 1-Myr mean bottom $\delta^{13}C$ trend between the composite record (Westerhold et al., 2020) and our simulations.

(Steinthorsdottir et al., 2021), which is close to our modeled long-term $pCO_2{}^{atm}$ change (~20% in our model). Nevertheless, despite inconsistencies that may still exist between the model output and proxy-based reconstructions, our results may have revealed the possible first-order long-term trend of the $pCO_2{}^{atm}$ change during the late Miocene.

Overall, the modeled sea water $\delta^{13}C$ in the INT experiment has shown good consistency with the late Miocene benthic and planktonic foraminiferal $\delta^{13}C$ records; and the modeled long-term trend of $pCO_2{}^{atm}$ in the INT experiment is well in line with that of the $pCO_2{}^{atm}$ reconstructions. Therefore, the model-data consistency is well established in the INT experiment, which forms the base to discuss the mechanisms driving the LMCIS.

4. Discussion

4.1. Mechanisms driving the LMCIS

In the INT experiment, one of the most essential features of the results is that the modeled $\delta^{13}C_{surface}$ and $\delta^{13}C_{bottom}$ show similar negative shift during the LMCIS, about 0.63‰ for $\delta^{13}C_{surface}$ and 0.72‰ for $\delta^{13}C_{bottom}$ (Fig. 4C, Fig. 5A). Our modeled results

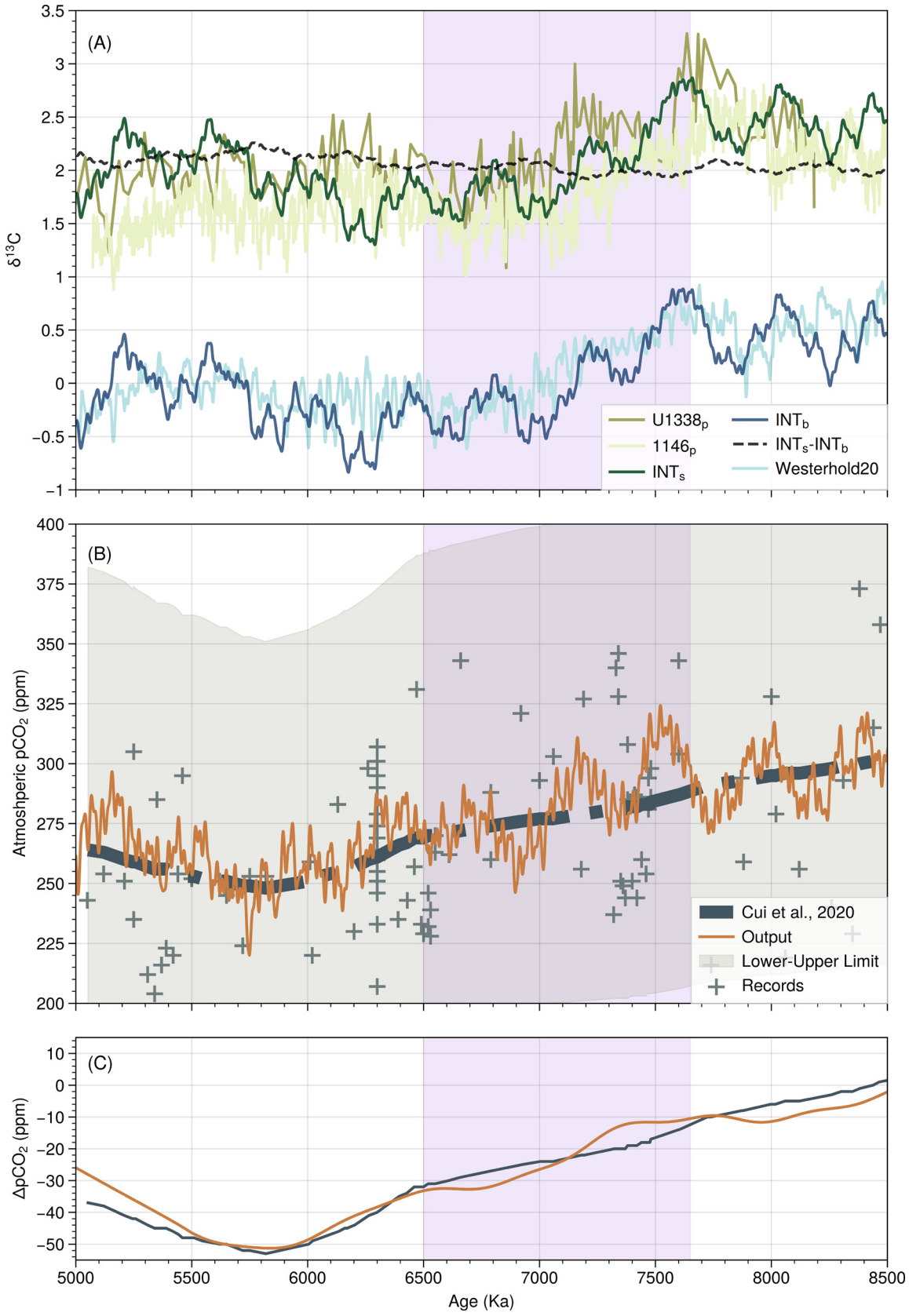


Fig. 5. Modeled $\delta^{13}C$ and pCO_2^{atm} in the INT experiment compared with reconstructions. (A) LMCIS downward the water column. $U1338_p$ and 1146_p refer to $\delta^{13}C_{planktonic}$ at IODP Site U1338 and ODP Site 1146, respectively. INT_s refers to modeled $\delta^{13}C_{surface}$ in the INT experiment, while INT_b refers to modeled $\delta^{13}C_{bottom}$. $INTs\text{-}INT_b$ refers to the modeled vertical $\delta^{13}C$ gradient. (B) Comparison of modeled pCO_2^{atm} in the INT experiment vs. pCO_2^{atm} reconstruction. In (B), all reconstruction data are from Cui et al., 2020. The thick black dashed line is the long-term mean of reconstructed pCO_2^{atm}, and the plus signs (+) are the paleo C_3-plant records used in their study. (C) Comparison of the long-term pCO_2^{atm} draw-down in the INT experiment (brown) and the reconstruction (black). The lavender columns in the subplots refer to the time interval of the LMCIS (~7.65 to 6.5 Ma).

are consistent with the benthic and planktonic foraminiferal $\delta^{13}C$ records of global oceans which display synchronously similar decreases during the LMCIS. The model-data consistency indicates that the LMCIS took place not only in the deep ocean but also in the upper ocean. Therefore, mechanisms which would substantially enlarge the vertical $\delta^{13}C$ gradient by transferring more ^{12}C into the deep ocean and leaving the upper ocean ^{13}C-enriched (e.g., biogenic bloom events) could not have been the primary driver of the LMCIS.

In the surface ocean, two main processes could influence $\delta^{13}C_{surface}$. First, the enhanced ^{13}C-depleted weathering input would lead to negative $\delta^{13}C$ shift via mixing. Second, associated with the enhanced weathering input was more nutrient (silicate and phosphate in our case) input, which would stimulate the biological pump, in favor of ^{13}C enrichment in the surface ocean because the photosynthesis tends to utilize ^{12}C first. The two processes affect $\delta^{13}C_{surface}$ in an opposite manner, but the role of surface mixing is dominant as the amount of excess weathering input ($\sim10^{12}$ mol C/yr) is $\sim$10 times larger than the surface production ($\sim10^{11}$ mol C/yr) in our model, leading to the long-term negative shift of $\delta^{13}C_{surface}$.

In the deep ocean, mechanisms driving the LMCIS are more complicated, as the ocean interior does not respond to the terrestrial forcing directly, which would have involved different processes. First, the thermohaline circulation defined in our model (Q1, Q2, Q3 in Fig. 1F) would gradually transfer the surface water containing more carbon with negative $\delta^{13}C$ into the deep ocean. Along the pathway of the prescribed thermohaline circulation, $\delta^{13}C_{bottom}$ would decrease due to local mixing. Second, as we fix the fractionation constant to be -23‰ during the photosynthesis, $\delta^{13}C$ of the organic matter from surface production would also decrease when $\delta^{13}C_{surface}$ decreases. The remineralization of sinking organic matter, therefore, would release more ^{12}C to the deep ocean in the INT experiment even if the surface production remains unchanged, resulting in the decrease of $\delta^{13}C_{bottom}$. Third, as mentioned above, surface production is in fact stimulated by more nutrient input in the INT experiment, which leads to more sinking organic particles, facilitating $\delta^{13}C_{bottom}$ decrease. In this way, the more the surface production is enhanced, the more ^{12}C is transferred into the deep ocean, leaving the upper ocean ^{13}C-enriched. Therefore, this process explains why the vertical $\delta^{13}C$ gradient in the INT experiment is slightly enlarged after the LMCIS.

4.2. Hypothesis: link between the LMCIS and tectonic uplift events

Based on the results in the sensitivity experiments, separated terrestrial process, either derived from the C_4-expansion (VEG) or the enhanced weathering input (LAND), does not result in the expected amplitude of the negative sea water $\delta^{13}C$ excursion. Only if the two terrestrial processes combined together (as in the INT experiment) would the modeled long-term $\delta^{13}C$ shift be consistent with the planktonic and benthic foraminiferal $\delta^{13}C$ records (Fig. 4C, Fig. 5A). If this is the case, the onset (and end) of the C_4-expansion and the enhanced weathering input should be synchronous to each other. This indicates that the two processes may be linked to the same triggering mechanism that could be in turn linked with the tectonic uplift (Fig. 6).

Before the LMCIS, the modeled long-term trend of $\delta^{13}C$ in the INT experiment shows no inclination of excursion (Fig. 4C; Fig. 5A), indicating a rather steady tectonic/climatic background (Fig. 6A). At this point, the land biome is dominated by the ^{13}C-depleted C_3 plants (Fig. 1D). The onset of the LMCIS at $\sim$7.6 Ma coincides with the intensive uplift events in the northeast Tibetan Plateau (Li et al., 2014; Miao et al., 2012) starting at $\sim$8 Ma (Fig. 6B), known as the Late Miocene Uplift of the Tibetan Plateau (LMUTP hereafter). The LMUTP exerted potent effects on the regional and global climate. First, model studies have shown that the LMUTP can reduce precipitation in central Asia via reorganization of large-scale atmospheric circulation (An et al., 2001; Liu et al., 2015; Zhang et al., 2015). The central Asia aridification during the late Miocene, supported by palynological evidence and clay mineralogical records (Liu et al., 2016; Yang et al., 2019), is evidently in favor of C_4-expansion, as C_4-grasses show better tolerance than C_3-forest to dry environment or strong seasonality (Osborne, 2008). Second, the LMUTP would have exposed larger surface areas, providing a large quantity of fresh rocks to be eroded by intensive physical and chemical weathering (Liu et al., 2015; Miao et al., 2012). In this way, the LMUTP would significantly have affected the land-sea carbon (isotopic) budget via intensification of weathering flux with much more ^{12}C due to the C_4-expansion as prescribed in the INT experiment. Following the processes described in Section 4.1., both the $\delta^{13}C_{surface}$ and $\delta^{13}C_{bottom}$ undergo drastic negative shift, i.e., the LMCIS. The $pCO_2{}^{atm}$ during the LMCIS tends to decrease due to the more alkaline ocean and the stimulated biological pump (discussed later in Section 4.3.). In addition, the demise of the LMCIS can also be linked to the end of the LMUTP (Fig. 6C). The excess fresh rock exposed to weathering due to LMUTP would be consumed without further significant uplift events (from 6 Ma on). This would result in increased weathering flux than that under pre-LMCIS climate (because of the enlarged exposed area), but it would be (partly) buffered by low $pCO_2{}^{atm}$ (Eq. S1). C_4-expansion would also end due to the environmental limit at the end of the late Miocene, stabilizing the $\delta^{13}C$ of land-sea flux. The cessation of terrestrial forcing would not support further excursion of seawater $\delta^{13}C$, marking the end of the LMCIS.

4.3. Long-term $pCO_2{}^{atm}$ interactions with the strength of weathering

The modeled secular change of $pCO_2{}^{atm}$ in the INT experiment decreases by $\sim$50 ppm from 8 to 5.8 Ma (Fig. 5B). As the C_4-expansion does not affect the carbon budget based on our model design (because the C_4-expansion only influences $\delta^{13}C_{land_sea}$, Eq. S10), the $pCO_2{}^{atm}$ draw-down should result from the enhanced weathering input. Based on our results, two processes contributed to the $pCO_2{}^{atm}$ drawdown. First, it is evident that riverine alkalinity input increases associated with enhanced chemical weathering. In the INT experiment (and also LAND), together with $\sim$2100 Gt more carbon being put into the ocean, around 1.2×10^{16} mol more alkalinity (by calculation, including carbonate and silicate) is put into the ocean. The excess alkalinity input, in turn, increases the ocean's capacity to hold carbon. Second, more nutrients ($\sim2.2\times10^{15}$ mol silicate and $\sim1.1\times10^{14}$ mol phosphate in our case) are put into the ocean in the INT experiment, lowering $pCO_2{}^{atm}$ by stimulating the surface biological pump. We note the numbers (excess alkalinity and nutrients input) are the accumulation of the riverine flux anomaly between the INT (and LAND) and the CNTR experiment (from 8 to 6 Ma) controlled by the ψ parameter (Eq. S1) but not a one-time perturbation as a change in weatherability.

Long-term $pCO_2{}^{atm}$ draw-down has been proposed the core mechanism for the Late Miocene Cooling via significant long-wave heat loss (Herbert et al., 2016). Based on this, our results reveal that the re-partitioning of carbon, alkalinity, and nutrient between the land-ocean system could be the main cause of the long-term $pCO_2{}^{atm}$ change. Specifically, long-term changes in the $pCO_2{}^{atm}$ and the weathering flux exhibit a negative feedback in the INT experiment. For instance, at the beginning of the LMCIS ($\sim$7.6 Ma), the modeled $pCO_2{}^{atm}$ is as high as 320 ppm (Fig. 5B), in favor of a strong weathering process (Eq. S1). The strong weathering flux transferred alkalinity and nutrients into the ocean, lowering the $pCO_2{}^{atm}$. Further, when the terrestrial forcing is ceased at $\sim$6 Ma, modeled $pCO_2{}^{atm}$ decreases to $\sim$250 ppm (Fig. 5B), facilitating a

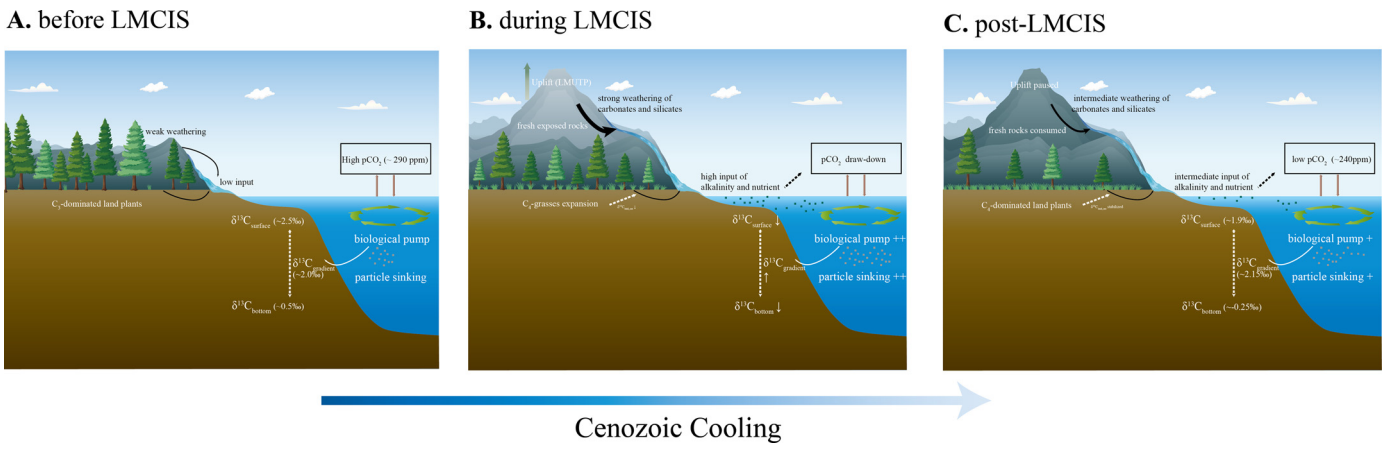


Fig. 6. Schematic illustrating the hypothesis we proposed in this study.

weaker weathering process (compared to the pre-LMCIS state). The ocean in the post-LMCIS condition hence receives less alkalinity and nutrient than that in the early-LMCIS condition, leading to a $pCO_2{}^{atm}$ increase starting from ~5.8 Ma.

Overall, our results show that the negative feedback between the long-term changes of the $pCO_2{}^{atm}$ and weathering fluxes tend to stabilize the Earth's surface system climate by lowering $pCO_2{}^{atm}$ when it is high and vice versa. In this case, $pCO_2{}^{atm}$ can be an amplifier in the climate system but is not necessarily the primary driver of the late Miocene climate change. In comparison, our results indicate that the tectonic perturbation (e.g., LMUTP) may be more important in determining the late Miocene carbon and hydrological cycle by causing the long-term $pCO_2{}^{atm}$ draw-down and further inducing large-scale reorganization of atmospheric circulation via orographic forcing, as is also shown in a recent modeling study (Farnsworth et al., 2019).

5. Concluding remarks and outlook

Based on a box-model simulation and model-data comparison, we reveal that the synergy between C_4-expansion and enhanced weathering input during the late Miocene, linked with the LMUTP, could ultimately result in the LMCIS of global ocean. The carbon-climate dynamics driven by a tectonic perturbation led to the re-partitioning of carbon among the terrestrial, oceanic, and atmospheric carbon reservoirs during the late Miocene.

We note, however, that further efforts are needed to elaborate on the late Miocene carbon cycle. For instance, the orographic forcing of LMUTP may result in a radical reorganization of oceanic meridional overturning circulation by re-distributing density flux at high-latitude Pacific and Atlantic sites (Su et al., 2018). This would influence the source-water $\delta^{13}C$ and inter-basin $\delta^{13}C$ gradients (Hodell and Kathryn, 2006) and should be further investigated in a revised model to better constrain the mechanisms driving the LMCIS.

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CRediT authorship contribution statement

J.T. led the project. J.D., J.T., and W.M. designed the experiments. J.D. performed the experiments. All authors contributed to interpreting the data and writing the paper.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All model outputs in this study have been deposited to the Pangaea repository (www.pangaea.de) and will be available for all readers after a specific Pangaea link is generated. Alternatively, readers can access the model outputs by sending a request to the authors (dujinlong@tongji.edu.cn) before the link is given. Code of the box-model (in Fortran) can be obtained from the authors by sending a request to dujinlong@tongji.edu.cn.

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Appendix A. Supplementary material

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.epsl.2022.117457.

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